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The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include 'simatslab.saveetha.com', 'SIMATS Online Lab', 'Internet Speed Test', 'Classwork for CSAD', 'CSAD925-JAVA PRO', and 'CSAD925/Lab Practi'. The URL bar shows 'simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=control'. The page header features the SIMATS ENGINEERING logo, the title 'SIMATS Online Programming Lab', and the user profile 'NITHISH RAAJU V' with ID '192411016RD'. The left sidebar shows 'QUESTIONS 31 / 31 completed' and a dropdown menu with 'Arithmetic' selected. Below it, the 'ARITHMETIC' section contains the instruction: 'Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.' The 'PROGRAM EDITOR' section has a 'Java' dropdown and 'Run' and 'Save' buttons. The code editor contains the following Java code:

```
1 public class arithmeticoperations{
2     public static void main(String[] args){
3         int a=10;
4         int b=30;
5         int sum=a+b;
6         int diff=a-b;
7         int mul=a*b;
8         int div=a/b;
9         System.out.println("addition of two numbers is :"+sum);
10        System.out.println("subtraction of two numbers is:"+diff);
11        System.out.println("multiplication of two numbers is:"+mul);
12        System.out.println("division of two numbers is :"+div);
13    }
14 }
15 }
```

The 'OUTPUT' section shows the results of the program execution:

```
addition of two numbers is :40
subtraction of two numbers is :-20
```

The 'INPUT' section on the right contains the text 'Your INPUT goes here!'.

The screenshot shows the SIMATS Online Programming Lab interface for a different question. The browser tabs and URL bar are the same as in the previous screenshot. The left sidebar shows 'QUESTIONS 31 / 31 completed' and a dropdown menu with 'MAX' selected. Below it, the 'MAX' section contains the instruction: 'Write a Java program to find the maximum of three numbers using conditional statements.' The 'PROGRAM EDITOR' section has a 'Java' dropdown and 'Run' and 'Save' buttons. The code editor contains the following Java code:

```
1 import java.util.Scanner;
2 public class max{
3     public static void main(String[] args){
4         int a,b,c;
5         int max;
6         Scanner ex = new Scanner(System.in);
7         a=ex.nextInt();
8         b=ex.nextInt();
9         c=ex.nextInt();
10
11         if (a>b && a>c)
12             max=a;
13         else if (b>c)
14             max=b;
15         else
16             max=c;
17         System.out.println("the maximum element is :"+max);
18
19
20 }
21 }
```

The 'OUTPUT' section shows the result of the program execution:

```
the maximum element is :5
```

The 'INPUT' section on the right contains the numbers '2', '5', and '3' on separate lines.

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QUESTIONS 31 / 31 completed

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class leap{
3     public static void main(String[] args){
4         int a;
5         Scanner ex= new Scanner(System.in);
6         a=ex.nextInt();
7         if (a%4==0)
8             System.out.println("leap year:"+a);
9         else
10            System.out.println("not a leap year:"+a);
11     }
12 }
```

INPUT

2002

OUTPUT

not a leap year:2002

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QUESTIONS 31 / 31 completed

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner ex=new Scanner(System.in);
5         int a=0,b=1,c;
6         System.out.println("enter how many terms :");
7         int n=ex.nextInt();
8         for (int i=1;i<=n;i++){
9             System.out.print(a+" ");
10            c=a+b;
11            a=b;
12            b=c;
13        }
14    }
15 }
```

INPUT

10

OUTPUT

enter how many terms :
0 1 1 2 3 5 8 13 21 34

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QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class factorial{
3     public static void main(String[] args){
4         Scanner ex=new Scanner(System.in);
5
6         System.out.println("enter the number :");
7         int a=ex.nextInt();
8         int c=1;
9         for (int i=1;i<=a;i++){
10             c=c*i;
11         }
12         System.out.println("the factorial of the entered number is :"+c);
13     }
14 }
15 }
16 }
```

INPUT

5

OUTPUT

enter the number :
the factorial of the entered number is :120

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QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class rt{
3     public static void main(String[] args){
4         Scanner ex=new Scanner(System.in);
5         System.out.println("enter how many rows:");
6         int a=ex.nextInt();
7         for (int i=1;i<=a;i++){
8             for (int j=1;j<=i;j++){
9                 System.out.print("* ");
10             }
11             System.out.println();
12         }
13     }
14 }
```

INPUT

5

OUTPUT

enter how many rows:
*
*
*
*
*

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QUESTIONS 31 / 31 completed

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class search{
3     public static void main(String[] args){
4         Scanner ex=new Scanner(System.in);
5         int a[]={10,20,30,40,50};
6         System.out.print("Enter the nnumber to search :");
7         int key=ex.nextInt();
8         int low=0;
9         int high=a.length-1;
10        int mid;
11        while (low<=high){
12            mid=(low+high)/2;
13            if (a[mid]== key){
14                System.out.println("Element foundat index "+mid);
15                return;
16            } else if (key<a[mid]){
17                high=mid-1;
18            } else{
19                low=mid+1;
20            }
21        }
```

OUTPUT

Enter the nnumber to search :Element not found

INPUT

2

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QUESTIONS 31 / 31 completed

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("enter two numbers:");
6         int a=sc.nextInt();
7         int b=sc.nextInt();
8         int gcd=1;
9         for (int i=1;i<=a && i<=b;i++){
10             if (a%i==0 && b%i==0)
11                 gcd=i;
12         }
13         System.out.println("gcd="+gcd);
14     }
15 }
```

OUTPUT

enter two numbers: gcd=6

INPUT

12 18

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QUESTIONS 31 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class prime{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("Enter a number :");
6         int n=sc.nextInt();
7         int count=0;
8         for (int i=1;i<=n;i++){
9             if (n%i==0){
10                 count++;
11             }
12         }
13         if (count==2){
14             System.out.println(n+"is a prime number ");
15         }else{
16             System.out.println(n+"it is not a prime number");
17         }
18     }
19 }
20 }
21 }
```

INPUT

12

OUTPUT

Enter a number :12it is not a prime number

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QUESTIONS 31 / 31 completed

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class selection{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("enter the number of elements:");
6         int n=sc.nextInt();
7         int[] a=new int[n];
8         System.out.println("enter the elements:");
9         for (int i=0;i<n;i++){
10             a[i]=sc.nextInt();
11         }
12         for (int i=0;i<n-1;i++){
13             int min=i;
14             for (int j=i+1;j<n;j++){
15                 if (a[j]<a[min]){
16                     min=j;
17                 }
18             }
19             int temp=a[i];
20             a[i]=a[min];
21             a[min]=temp;
22         }
23     }
24 }
```

INPUT

5
65
54
65
34
23

OUTPUT

enter the number of elements:enter the elements:
sorted array:

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QUESTIONS 31 / 31 completed

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class multiplication{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("enter a number:");
6         int n=sc.nextInt();
7         System.out.println("multiplication table of"+n);
8         for (int i=1;i<=10;i++){
9             System.out.println(n+"x"+i+"="+n*i);
10        }
11    }
12 }
```

INPUT

5

OUTPUT

enter a number:multiplication table of 5
5x1=5

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QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     static int fact(int n){
4         if (n==0||n==1)
5             return 1;
6         else
7             return n*fact(n-1);
8     }
9     public static void main(String[] args){
10        Scanner sc=new Scanner(System.in);
11        System.out.print("enter a number:");
12        int n=sc.nextInt();
13        System.out.println("factorial="+fact(n));
14    }
15 }
```

INPUT

5

OUTPUT

enter a number:factorial=120

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QUESTIONS 31 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****

```
*****
****
***
**
*
```

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class lt{
3     public static void main(String[] args){
4         Scanner ex=new Scanner(System.in);
5         System.out.print("enter no of rows:");
6         int a=ex.nextInt();
7         for (int i=a;i>=1;i--){
8             for (int j=1;j<=i;j++){
9                 System.out.print("*");
10            }
11        }
12        System.out.println();
13    }
14 }
15 }
```

INPUT

5

OUTPUT

```
enter no of rows:*****
*****
```

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QUESTIONS 31 / 31 completed

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

```
*****
****
***
**
*
```

PROGRAM EDITOR Java Run Save

```
1 public class lcm{
2     public static void main(String[] args){
3         int a=12;
4         int b=10;
5         int max;
6         if (a>b){
7             max=a;
8         }
9         else {
10            max=b;
11        }
12    }
13    while (true){
14        if (max%a==0 && max%b==0){
15            System.out.println("lcm =" +max);
16            break;
17        }
18        max++;
19    }
20 }
21 }
```

INPUT

Your INPUT goes here!

OUTPUT

```
lcm =60
```

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QUESTIONS 31 / 31 completed

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 public class Main{
2     public static void main(String[] args){
3         int n=13;
4         int binary=0;
5         int place=1;
6         while (n>0){
7             int rem=n%2;
8             binary=binary+(rem*place);
9             place=place*10;
10            n=n/2;
11        }
12        System.out.println("binary="+binary);
13    }
14 }
```

OUTPUT

binary=1101

INPUT

Your INPUT goes here!

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QUESTIONS 31 / 31 completed

B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 public class Main{
2     public static void main(String[] args){
3         int binary=1101;
4         int decimal=0;
5         int base=1;
6         while (binary>0){
7             int digit=binary%10;
8             if (digit==1){
9                 decimal=decimal+base;
10            }
11            base=base*2;
12            binary=binary/10;
13        }
14        System.out.println("decimal="+decimal);
15    }
16 }
17 }
18 }
```

OUTPUT

decimal=13

INPUT

Your INPUT goes here!

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QUESTIONS 31 / 31 completed

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 public class Main{
2     public static void main(String[] args){
3         int num=12345;
4         int sum=0;
5         while (num>0){
6             int digit=num%10;
7             if (digit>=0){
8                 sum=sum+digit;
9             }
10            num=num/10;
11        }
12        System.out.println("sum of digits="+sum);
13    }
14 }
```

OUTPUT

```
sum of digits=15
```

INPUT

Your INPUT goes here!

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QUESTIONS 31 / 31 completed

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 public class Main{
2     public static void main(String[] args){
3         int a=122345670;
4         int pro=1;
5         while (a>0){
6             int digit=a%10;
7             if (digit>=0){
8                 pro=pro*digit;
9             }
10            a=a/10;
11        }
12        System.out.println("product of numbers =" +pro);
13    }
14 }
```

OUTPUT

```
product of numbers =0
```

INPUT

Your INPUT goes here!

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QUESTIONS 31 / 31 completed

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 public class Main{
2     public static void main(String[] args){
3         int base=2;
4         int exponent=5;
5         int power=1;
6         if (exponent==0){
7             for (int i=1;i<=exponent;i++){
8                 power=power*base;
9             }
10        }
11        System.out.println("power="+power);
12    }else{
13        System.out.println("Exponent should be non negative.");
14    }
15 }
16 }
```

OUTPUT

power=32

INPUT

Your INPUT goes here!

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QUESTIONS 31 / 31 completed

Area Circle

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner ex=new Scanner(System.in);
5         int a=ex.nextInt();
6         double b=3.14;
7         System.out.println("the are of the circle is :"+(a*b*b));
8     }
9 }
10 }
```

OUTPUT

the are of the circle is :49.298

INPUT

5

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QUESTIONS 31 / 31 completed

Shpere

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("enter the radius:");
6         double radius=sc.nextDouble();
7         double volume;
8         volume=(4.0/3.0)*3.14159*radius*radius*radius;
9         System.out.println("volume of sphere="+volume);
10    }
11 }
```

INPUT

7

OUTPUT

enter the radius:volume of sphere=1436.7538266666666

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QUESTIONS 31 / 31 completed

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("enter a string:");
6         String str=sc.nextLine();
7         String rev="";
8         for (int i=str.length()-1;i>=0;i--){
9             rev=rev+str.charAt(i);
10        }
11    }
12    if (str.equals(rev)){
13        System.out.println("pallindrome");
14    }
15    }else {
16        System.out.println("not a pallindrome");
17    }
18 }
19 }
```

INPUT

malayalam

OUTPUT

enter a string:pallindrome

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QUESTIONS 31 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("enter a number:");
6         int num=sc.nextInt();
7         int original=num;
8         int sum=0;
9         while (num>0){
10             int digit=num%10;
11             sum=sum+(digit*digit*digit);
12             num=num/10;
13         }
14         if (sum==original){
15             System.out.println("armstrong number ");
16         }else{
17             System.out.println("not an armstrong number");
18         }
19     }
20 }
21 }
```

OUTPUT

enter a number:armstrong number

INPUT

153

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QUESTIONS 31 / 31 completed

RT-num.Pattern

**RT-
NUM.PATTERN**

Write a Java program to print the following pattern using loops:1
22
333
4444
55555

PROGRAM EDITOR Java Run Save

```
1 public class Main{
2     public static void main(String[] args){
3         for (int i=1;i<=5;i++){
4             for(int j=1;j<=i;j++){
5                 System.out.print(i);
6             }
7             System.out.println();
8         }
9     }
10 }
```

OUTPUT

1
22

INPUT

Your INPUT goes here!

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QUESTIONS 31 / 31 completed

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("enter a number");
6         int num=sc.nextInt();
7         int sum=0;
8         for (int i=1;i<num;i++){
9             if (num%i==0){
10                 sum=sum+i;
11             }
12         }
13         if (sum==num){
14             System.out.println("perfect number ");
15         }else{
16             System.out.println("neot a perfect number:");
17         }
18     }
19 }
```

INPUT

6

OUTPUT

enter a numberperfect number

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QUESTIONS 31 / 31 completed

Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("enter size:");
6         int n=sc.nextInt();
7         int [] a=new int[n];
8         System.out.println("enter elements:");
9         for(int i=0;i<n;i++){
10             a[i]=sc.nextInt();
11         }
12         int largest=a[0];
13         for (int i=1;i<n;i++){
14             if (a[i]>largest)
15                 largest=a[i];
16         }
17         System.out.println("largest element="+largest);
18     }
19 }
```

INPUT

5
10
24
67
78
67

OUTPUT

enter size:enter elements:
largest element=78

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QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("enter size:");
6         int n=sc.nextInt();
7         int[] a=new int[n];
8         System.out.println("enter elements:");
9         for (int i=0;i<n;i++){
10             a[i]=sc.nextInt();
11         }
12         for (int i=0;i<n-1;i++){
13             for (int j=i+1;j<n;j++){
14                 if (a[i]<a[j]){
15                     int temp=a[i];
16                     a[i]=a[j];
17                     a[j]=temp;
18                 }
19             }
20         }
21         System.out.println("second largest element="+a[1]);
22     }
23 }
```

INPUT

5
10
25
8
40
30

OUTPUT

enter size:enter elements:
second largest element=30

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QUESTIONS 31 / 31 completed

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("enter size:");
6         int n=sc.nextInt();
7         int[] a=new int[n];
8         System.out.println("enter elements:");
9         for(int i=0;i<n;i++){
10             a[i]=sc.nextInt();
11         }
12         int smallest=a[0];
13         for (int i=1;i<n;i++){
14             if(a[i]<smallest)
15                 smallest=a[i];
16         }
17         System.out.println("smallest element="+smallest);
18     }
19 }
```

INPUT

5
39
89
384
876
12

OUTPUT

enter size:enter elements:
smallest element=12

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JAVA PROGRAMMING



QUESTIONS 31 / 31 completed

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main{
2     public static void main(String[] args){
3         int[] a={5,4,8,1,4};
4         int n=a.length;
5         for (int i=1;i<n;i++){
6             int key=a[i];
7             int j=i-1;
8             while (j>=0 && a[j]>key){
9                 a[j+1]=a[j];
10                j--;
11            }
12            a[j+1]=key;
13        }
14        System.out.println("sorted array:");
15        for (int i=0;i<n;i++){
16            System.out.print(a[i]+" ");
17        }
18    }
19 }
20 }
```

OUTPUT

sorted array:
1 4 4 5 8

INPUT

Your INPUT goes here!